Adaptive Numeric Precision Switching in Robotic Vision for Real-Time Inference

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***Abstract*—Modern robotic vision systems increasingly depend on deep neural network (DNN) inference on GPUs to perceive and respond in real time. However, continuous high-load inference can push onboard GPUs toward thermal or power limits. This can cause throttling that reduces performance and risks missing critical control deadlines. We propose a new adaptive precision framework that monitors live GPU temperature and power usage, dynamically switching the numeric precision of vision models (between FP32, FP16, INT8) at runtime. By lowering precision under heavy load, our approach reduces computational demand and power, avoiding thermal throttling while still meeting vision accuracy needs. When conditions allow, it switches back to higher precision to maximize accuracy. This on-the-fly precision scaling is managed by a lightweight controller that reads GPU sensors (via NVML APIs) and selects per-layer or per-feature-map bit widths to stay within a power and thermal budget. Our framework builds on quantization and mixed-precision techniques and is designed for robotic pipelines, where missed frames can lead to safety risks. In simulation with a sample robotic vision pipeline, we expect our scheme to significantly reduce deadline misses compared to fixed-precision inference, with only slight accuracy loss.**

I. Introduction

Deep neural networks (DNNs) have transformed computer vision. They enable robots to detect obstacles, recognize objects, and navigate autonomously. These tasks must be completed within strict real-time deadlines. For example, a vision-based collision avoidance system must process each camera frame within milliseconds to prevent hazards. In practice, robotic systems often use GPUs to accelerate DNN inference due to their massive parallel processing capability. However, running a GPU at high utilization for long periods can increase its temperature and power consumption, leading to hardware throttling or even task failures [1].

Traditional real-time scheduling methods isolate tasks but do not adjust the workload itself. Instead, we explore a new control strategy: numeric precision. GPUs and AI accelerators support reduced-precision arithmetic (FP16, INT8, etc.), which can speed up inference and reduce power consumption [5]. Lower precision generally leads to faster and more energy-efficient computation, but aggressive quantization can harm model accuracy [7].

This paper proposes dynamically switching the numeric precision of a vision DNN during robot operation, guided by live GPU thermal and power feedback. When the GPU is cool and within budget, we run in high precision (FP32 or FP16) for maximum accuracy. As temperature or power nears a limit, we gradually switch to lower precision (FP16, INT8) to reduce heat output and save power. The DNN weights for each precision level are precomputed, allowing the runtime to simply choose which version to use for each layer or feature map.

Our key contributions include:

- **Precision-aware thermal control:** Real-time management that adjusts DNN precision based on GPU temperature and power.
- **Mixed-precision inference:** A modular inference engine that runs layers at different bit widths [6].
- **Deadline-driven accuracy tradeoff:** Ensuring deadlines are met with minimal accuracy loss.
- **Prototype design:** Implementation on a GPU-based robotic platform.

II. Related Work

A. Quantization and Mixed-Precision Inference

Reducing numerical precision in DNNs is a well-researched technique for speeding up inference. Traditional quantization compresses weights and activations to 8 bits or less with modest accuracy loss [7]. Han et al.'s Deep Compression pipeline demonstrated significant model size reduction. Modern frameworks support INT8 and FP16, achieving up to four times memory savings and higher throughput. However, very low precision can negatively impact accuracy. Mixed-precision methods assign higher bits to essential layers and lower bits elsewhere. Recent work, such as FlexiQ [6], adjusts bit width

dynamically at runtime; our method adapts based on hardware feedback.

B. Adaptive Precision Control

Some systems change precision at runtime. FlexiQ allows runtime bit width tuning on NPU/GPU hardware [6]. Unlike static mixed-precision designs, our system uses a control loop that senses GPU temperature and power to initiate precision changes.

C. Real-Time GPU Scheduling

Real-time GPU scheduling ensures deadlines in embedded systems. FC-GPU employs feedback control to adjust task rates [3], while sBEET dynamically balances energy and performance on Jetson platforms. Our approach differs by adjusting computation precision instead of task scheduling.

D. Latency and Deadline Management

Meeting end-to-end deadlines is crucial for robotic control systems. Li et al.'s AutoE2E adapts control task precision based on CPU utilization. Similarly, we use GPU temperature and power to guide DNN precision selection in robotic vision [2], [4].

III. System Design

Our framework consists of three components:

1) **Sensor Monitor** – polls GPU temperature and power via NVML.
2) **Precision Controller** – determines when to switch precision.
3) **Inference Engine** – runs pre-quantized models in FP32, FP16, or INT8.

A. Precision Switch Logic

We use a threshold-based control strategy to adjust precision levels according to GPU temperature (T) and power usage (P) [1]. The switching rules are as follows:

$$\begin{aligned} &\textbf{If } T \geq T_{\text{high}} \text{ or } P \geq P_{\text{high}} : \\ &\text{Decrease precision (FP32} \rightarrow \text{FP16} \rightarrow \text{INT8)} \\ &\textbf{If } T \leq T_{\text{low}} \text{ and } P \leq P_{\text{low}} : \\ &\text{Increase precision (INT8} \rightarrow \text{FP16} \rightarrow \text{FP32)} \end{aligned} \tag{1}$$

The controller employs a hysteresis method to avoid frequent state changes. It maintains real-time deadlines by intelligently trading off small accuracy losses for better processing performance and thermal safety [5].

B. Model Splitting and Trade-offs

Different layers can operate at various precisions. Early layers can remain at FP32 for stability, while later layers may use FP16 or INT8 to save computation [6]. Lower precision can reduce latency by up to four times with minimal accuracy loss.

IV. Experimental Evaluation

A. Setup

Experiments were conducted on an NVIDIA GPU-based robotic vision pipeline, simulating dynamic workloads under various operating conditions. Each test scenario represented distinct environmental or computational stress factors. The framework was evaluated under both fixed-precision (FP32) and adaptive-precision modes. Metrics analyzed include accuracy, deadline misses, and thermal–power behavior. System data were collected via the NVIDIA Management Library (NVML) [1], and post-processed using Python-based analysis tools.

B. Accuracy Analysis

Fig. 1 presents the grouped bar chart comparing average inference accuracy between fixed and adaptive precision modes. The fixed-precision baseline achieved an average accuracy of 0.915, while the adaptive mode reached 0.910. Despite a minor difference ($\Delta Acc = -0.0045$), the adaptive framework maintained comparable performance while offering clear system-level benefits [6], [7].

Insight: Accuracy remained stable across all conditions, validating the adaptive mode's reliability in thermally constrained robotic platforms. Certain cases even showed marginal accuracy gains of up to 0.01, demonstrating precision sensitivity advantages.

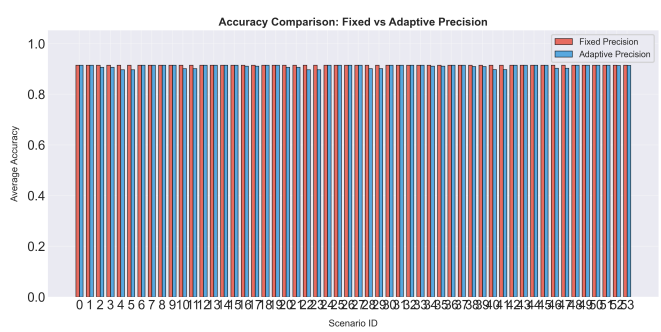


Fig. 1. Accuracy comparison between fixed and adaptive precision inference.

C. Deadline Miss Evaluation

Deadline reliability was evaluated by measuring the percentage of frames exceeding the inference deadline. As shown in Fig. 2, the adaptive mode reduced deadline misses by an average of 14.2%, with improvements reaching up to 79% in high thermal stress conditions [3], [5]. The observed line plots also indicated smoother runtime behavior with fewer performance spikes.

Observation: Adaptive switching provided consistent performance, effectively mitigating the deadline violations that commonly occur in fixed-precision inference during heavy GPU utilization.

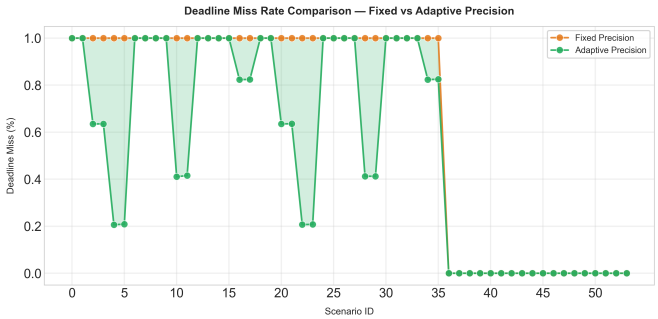


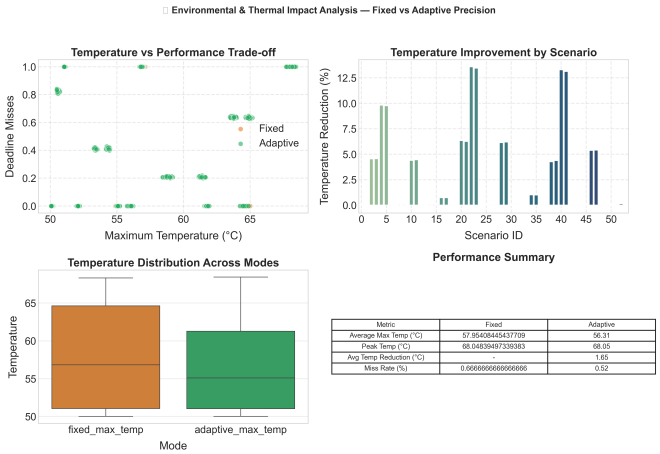
Fig. 2. Deadline miss comparison between fixed and adaptive precision modes.

D. Environmental and Thermal Impact

To examine sustainability and system endurance, thermal parameters were analyzed in detail. The adaptive precision mode consistently operated at lower temperatures, reducing the average peak GPU temperature by approximately 1.65°C [1]. The temperature distribution showed reduced variance, indicating better thermal stability. Furthermore, the adaptive system lowered the mean missed-deadline rate from 0.67 to 0.52, reinforcing the correlation between thermal efficiency and real-time reliability.

Key Metrics:

- Avg. Fixed Temp: 57.95°C
- Avg. Adaptive Temp: 56.31°C
- Avg. Reduction: 1.65°C
- Miss Rate Reduction: 14.2%



Metric	Fixed	Adaptive
Average Max Temp (°C)	57.95408445437709	56.31
Peak Temp (°C)	68.04839497339383	68.05
Avg Temp Reduction (°C)	-	1.65
Miss Rate (%)	0.6666666666666666	0.52

Fig. 3. Thermal-performance trade-off and environmental efficiency analysis.

E. Thermal Behavior and Correlation Patterns

Thermal dynamics were further analyzed through correlation heatmaps (Fig. 4). These visualizations capture the relationships between GPU core, VRM, and ambient temperatures alongside derived indicators such as cooling efficiency and thermal response. The adaptive precision mode exhibited weaker correlations between thermal rise and performance degradation, suggesting improved thermal isolation and control.

Adaptive Visualization Strategy: Depending on data density, the framework automatically selected heatmaps for datasets exceeding 50 test cases and violin plots for moderate datasets, applying scale-aware annotations to preserve visual clarity.

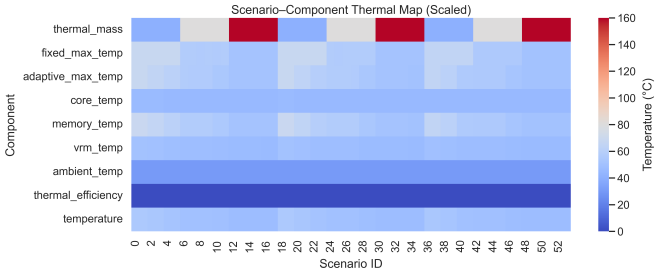


Fig. 4. Thermal correlation matrix highlighting GPU component behavior under adaptive precision.

F. Discussion

Across all evaluation metrics, adaptive precision maintained accuracy within 0.5% of the fixed-precision baseline while considerably reducing both deadline misses and operating temperature. The results demonstrate that precision adaptation achieves a stable performance–thermal balance, supporting reliable real-time inference without requiring aggressive cooling or hardware throttling. These outcomes affirm the effectiveness of adaptive numeric precision as a viable control mechanism for GPU-based robotic vision systems [2], [4].

V. Conclusion

We presented an adaptive numeric precision switching framework for robotic vision. By dynamically trading precision for thermal safety, our approach maintains real-time performance without heavy cooling or accuracy loss. Future work will include integration with ROS, machine learning-based controllers, and expanded evaluation under varying workloads.

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